Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

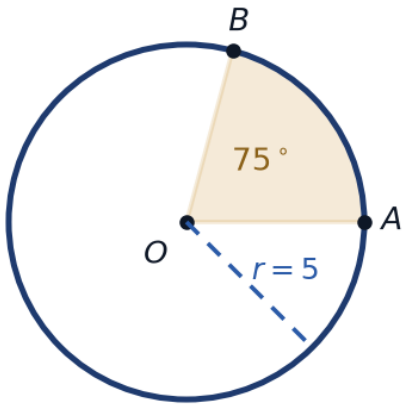
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The unit circle and radian measure

Convert 210° to radians.

2 [Foundational] • Target: The unit circle and radian measure

Convert $\frac{5\pi}{4}$ radians to degrees.



3 [Foundational] • Target: The unit circle and radian measure

Find the exact value of $\sin(240^\circ)$.

4 [Foundational] • Target: Trigonometric identities

Simplify $\frac{1-\sin^2\theta}{\cos\theta}$.

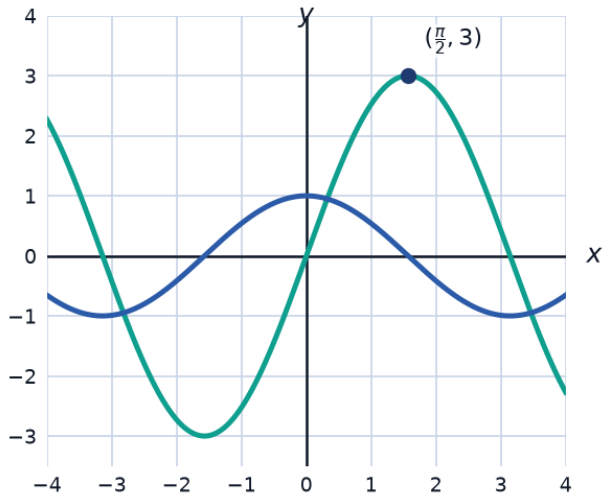
5 [Foundational] • Target: Trigonometric identities

Simplify $\tan\theta \cdot \cos\theta$.

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6 [Foundational] • Target: Trigonometric identities

If $\sin\theta = \frac{3}{5}$ and θ is in Quadrant II, find $\cos\theta$ and $\tan\theta$.



7 [Foundational] • Target: Double angle formulas

Given $\cos\theta = \frac{5}{13}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Double angle formulas

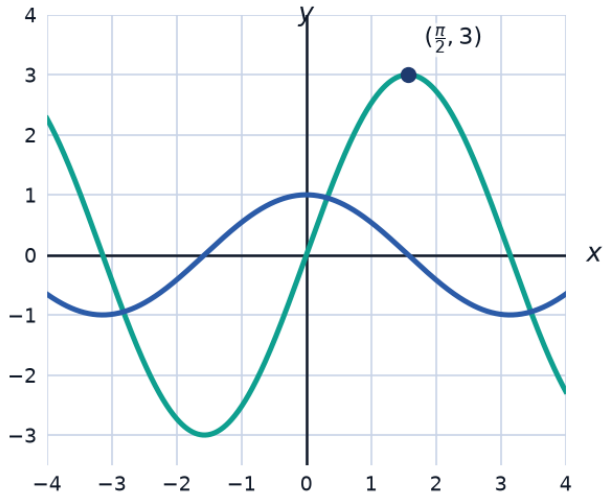
Use double-angle formulas to express $\sin 4\theta$ in terms of $\sin\theta$ and $\cos\theta$ only.

9 [Intermediate] • Target: Solving trigonometric equations

Solve $2\sin\theta - 1 = 0$ for $0^\circ \le \theta < 360^\circ$.

10 [Intermediate] • Target: Solving trigonometric equations

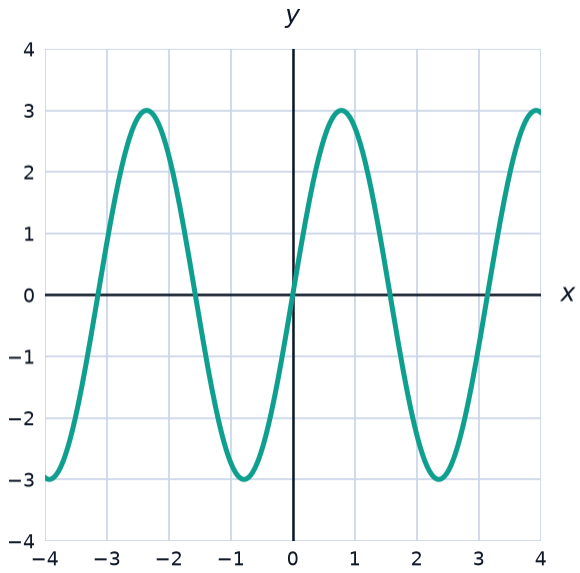
Solve $2\cos^2\theta + \cos\theta - 1 = 0$ for $0^\circ \le \theta < 360^\circ$.



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11 **[Intermediate]** • **Target: Graphs of trigonometric functions**

The graph shows $y = 3\sin(2x)$. State the amplitude and period.



12 **[Intermediate]** • **Target: Graphs of trigonometric functions**

The function $y = 4\cos(3x - \pi)$ has what amplitude, period, and phase shift?

13 **[Intermediate]** • **Target: The Law of Sines**

In triangle ABC, $a = 10$, $A = 40^\circ$, $B = 65^\circ$. Find b, to the nearest hundredth.

14 **[Intermediate]** • **Target: The Law of Sines**

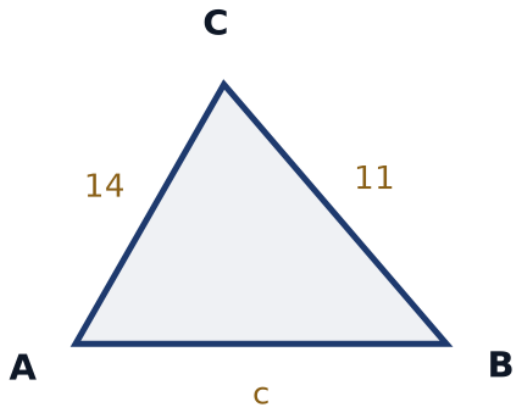
In triangle ABC, $a = 8$, $b = 10$, $A = 35^\circ$. Determine how many triangles are possible, and find angle B (to the nearest tenth of a degree) for each.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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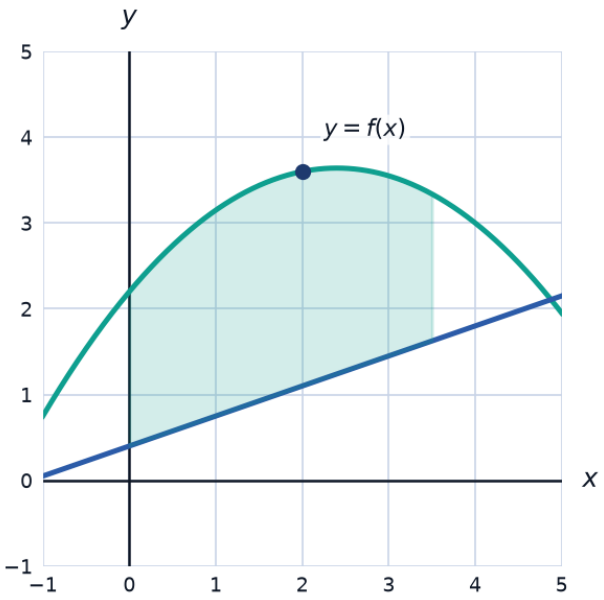
15 [Challenging] • Target: The Law of Cosines and the sine area formula • Cross-Lesson: 6.01 Circles & Coordinate Geometry

In triangle ABC shown below, $a = 11$, $b = 14$, and $C = 55^\circ$. Find side c, to the nearest hundredth, and the area of the triangle, to the nearest tenth.



16 [Challenging] • Target: The Law of Cosines and the sine area formula • Cross-Lesson: 7.01 Solid Geometry & Similarity

State the two formulas used above (Law of Cosines for a side, and the sine formula for area) in general form for a triangle with sides a, b, c and included angle C.



17 [Challenging] • Target: Inverse trigonometric functions • Cross-Lesson: 8.01 Statistics & Probability

Evaluate $\arcsin\left(\frac{\sqrt{2}}{2}\right)$ exactly, in radians.

18 [Challenging] • Target: Inverse trigonometric functions • Cross-Lesson: 1.01 Advanced Algebra

Evaluate $\arccos(-1)$ exactly, in radians.

19 [Challenging] • Target: Inverse trigonometric functions • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Find $\theta = \arctan(1)$ in degrees (principal value).

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20 [Challenging] • Target: A well-designed word problem: locating a ship from two lighthouses • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

This question has three parts. Two lighthouses A and B are 15 km apart along a straight coastline. A ship at point C is sighted such that $\angle CAB = 72^\circ$ and $\angle CBA = 58^\circ$ (both measured from line AB toward the ship). (a) Find $\angle ACB$. (b) Use the Law of Sines to find the distance AC from lighthouse A to the ship, to the nearest tenth of a km. (c) Find the distance BC from lighthouse B to the ship, to the nearest tenth of a km.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The unit circle and radian measure

Convert 135° to radians.

a. $\frac{3\pi}{5}$

b. $\frac{3\pi}{2}$

c. $\frac{3\pi}{4}$

d. π

22 [Foundational] • Target: Trigonometric identities

Convert 345° to radians.

a. 2π

b. $\frac{23\pi}{13}$

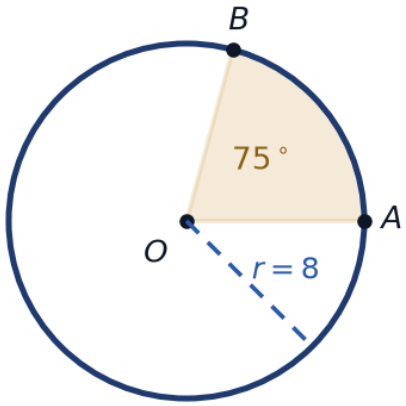
c. $\frac{23\pi}{12}$

d. $\frac{23\pi}{6}$

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23 [Foundational] • Target: Double angle formulas

Convert 90° to radians.



a. π

b. $\frac{3\pi}{2}$

c. $\frac{\pi}{3}$

d. $\frac{\pi}{2}$

24 [Foundational] • Target: Solving trigonometric equations

Convert 315° to radians.

a. $\frac{7\pi}{5}$

b. $\frac{7\pi}{2}$

c. 2π

d. $\frac{7\pi}{4}$

25 [Foundational] • Target: Graphs of trigonometric functions

Convert 165° to radians.

a. $\frac{11\pi}{6}$

b. $\frac{11\pi}{12}$

c. π

d. $\frac{11\pi}{13}$

26 [Foundational] • Target: The Law of Sines

Convert 15° to radians.

a. $\frac{\pi}{6}$

b. $\frac{\pi}{13}$

c. $\frac{\pi}{4}$

d. $\frac{\pi}{12}$

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27 [Foundational] • Target: The Law of Cosines and the sine area formula

Convert 240° to radians.

a. $\frac{5\pi}{3}$

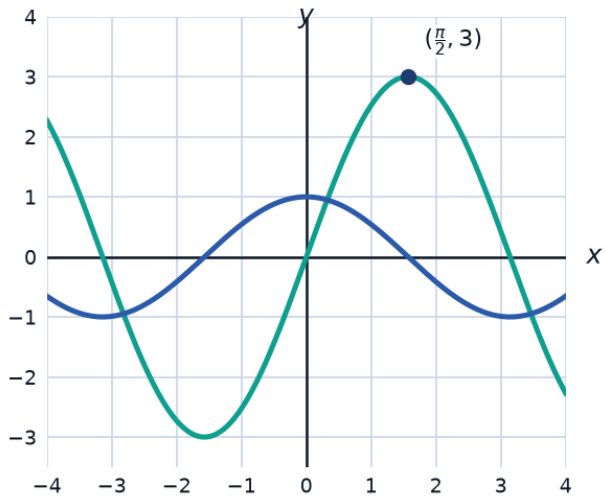
b. $\frac{4\pi}{3}$

c. π

d. $\frac{8\pi}{3}$

28 [Foundational] • Target: Inverse trigonometric functions

Solve $\tan\theta = -\frac{\sqrt{3}}{3}$ for $0^\circ \le \theta < 360^\circ$.



a. $30^\circ, 210^\circ$

b. $150^\circ, 330^\circ$

c. $135^\circ, 315^\circ$

d. $120^\circ, 300^\circ$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: locating a ship from two lighthouses

Solve $\sin\theta = -\frac{1}{2}$ for $0^\circ \le \theta < 360^\circ$.

a. $225^\circ, 315^\circ$

b. $210^\circ, 330^\circ$

c. $30^\circ, 150^\circ$

d. $45^\circ, 135^\circ$

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30 [Intermediate] • Target: Trigonometric Identities & Simplifications

Solve $\cos\theta = \frac{\sqrt{3}}{2}$ for $0^\circ \le \theta < 360^\circ$.

a. $135^\circ, 225^\circ$

b. $30^\circ, 330^\circ$

c. $120^\circ, 240^\circ$

d. $60^\circ, 300^\circ$

31 [Intermediate] • Target: Solving Trigonometric Equations

Solve $\sin\theta = \frac{\sqrt{3}}{2}$ for $0^\circ \le \theta < 360^\circ$.

a. $30^\circ, 150^\circ$

b. $240^\circ, 300^\circ$

c. $60^\circ, 120^\circ$

d. $210^\circ, 330^\circ$

32 [Intermediate] • Target: Laws of Sines & Cosines with Ambiguous Case

Solve $\tan\theta = \frac{\sqrt{3}}{3}$ for $0^\circ \le \theta < 360^\circ$.

a. $120^\circ, 300^\circ$

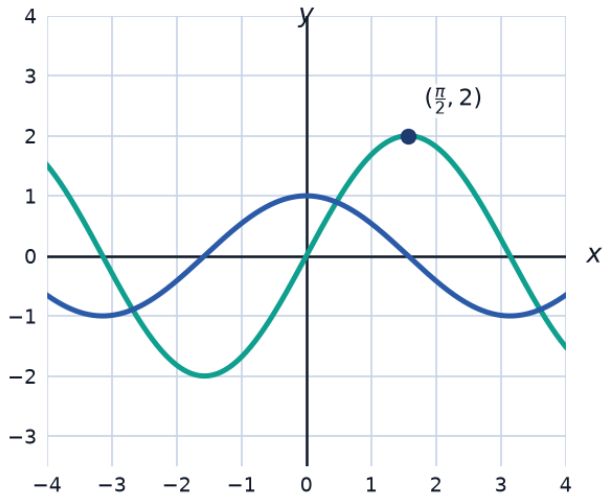
b. $30^\circ, 210^\circ$

c. $135^\circ, 315^\circ$

d. $60^\circ, 240^\circ$

33 [Intermediate] • Target: Periodic Modeling & Phase Shifts

Solve $\sin\theta = -\frac{\sqrt{3}}{2}$ for $0^\circ \le \theta < 360^\circ$.



a. $210^\circ, 330^\circ$

b. $240^\circ, 300^\circ$

c. $225^\circ, 315^\circ$

d. $60^\circ, 120^\circ$

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34 [Intermediate] • Target: The unit circle and radian measure

In triangle ABC, $a = 18$, $A = 80^\circ$, $B = 70^\circ$. Find b, to the nearest hundredth.

a. 17.68

b. 18.18

c. 17.18

d. 16.68

35 [Intermediate] • Target: Trigonometric identities

In triangle ABC, $a = 15$, $A = 60^\circ$, $B = 30^\circ$. Find b, to the nearest hundredth.

a. 9.16

b. 8.66

c. 8.16

d. 9.66

36 [Intermediate] • Target: Double angle formulas

In triangle ABC, $a = 17$, $A = 80^\circ$, $B = 45^\circ$. Find b, to the nearest hundredth.

a. 12.71

b. 12.21

c. 13.21

d. 11.71

37 [Intermediate] • Target: Solving trigonometric equations

In triangle ABC, $a = 13$, $b = 11$, and $C = 130^\circ$. Find side c, to the nearest hundredth, and the area, to the nearest tenth.

a. $21.77, 56.8$

b. $21.77, 54.8$

c. $20.77, 52.8$

d. $22.77, 54.8$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Graphs of trigonometric functions • Cross-Lesson: 1.01 Advanced Algebra

In triangle ABC, $a = 9$, $b = 12$, and $C = 70^\circ$. Find side c, to the nearest hundredth, and the area, to the nearest tenth.

a. $12.29, 52.7$

b. $11.29, 48.7$

c. $13.29, 50.7$

d. $12.29, 50.7$

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39 [Challenging] • Target: The Law of Sines • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Given $\cos\theta = \frac{24}{25}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

a. $-\frac{336}{625}, \frac{527}{625}$

b. $\frac{336}{625}, \frac{527}{625}$

c. $\frac{336}{625}, -\frac{527}{625}$

d. $\frac{527}{625}, \frac{336}{625}$

40 [Challenging] • Target: The Law of Cosines and the sine area formula • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Given $\cos\theta = \frac{12}{13}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

a. $-\frac{120}{169}, \frac{119}{169}$

b. $\frac{120}{169}, -\frac{119}{169}$

c. $\frac{119}{169}, \frac{120}{169}$

d. $\frac{120}{169}, \frac{119}{169}$

41 [Challenging] • Target: Trigonometric Identities & Simplifications • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Simplify the trigonometric expression: $\frac{1}{1+\sin x} + \frac{1}{1-\sin x}$.

a. $2\sec^2 x$

b. $2\csc^2 x$

c. $2\cos^2 x$

d. 1

42 [Challenging] • Target: Solving Trigonometric Equations • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Find all solutions to $2\cos^2 x + \cos x - 1 = 0$ in $[0, 2\pi)$.

a. $\frac{\pi}{3}, \frac{5\pi}{3}, \pi$

b. $\frac{\pi}{6}, \frac{11\pi}{6}, \pi$

c. $\frac{2\pi}{3}, \frac{4\pi}{3}$

d. $\frac{\pi}{4}, \frac{7\pi}{4}$

43 [Challenging] • Target: Solving Trigonometric Equations • Cross-Lesson: 7.01 Solid Geometry & Similarity

Solve $\tan^2 x = 3$ for $x \in [0, \pi]$.

a. $\frac{\pi}{3}, \frac{2\pi}{3}$

b. $\frac{\pi}{6}, \frac{5\pi}{6}$

c. $\frac{\pi}{3}$ only

d. $\frac{\pi}{4}, \frac{3\pi}{4}$

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44 [Challenging] • Target: Laws of Sines & Cosines with Ambiguous Case • Cross-Lesson: 8.01 Statistics & Probability

In ΔABC, $a = 5, b = 7,$ and $c = 8$. What is the measure of angle C?

a. 45°

b. 60°

c. 75°

d. 90°

45 [Challenging] • Target: Periodic Modeling & Phase Shifts • Cross-Lesson: 1.01 Advanced Algebra

What is the phase shift of the sinusoidal wave $y = 3\sin(2x - \frac{\pi}{3}) + 4$?

a. $\frac{\pi}{6}$ to the right

b. $\frac{\pi}{3}$ to the right

c. $\frac{\pi}{6}$ to the left

d. $\frac{\pi}{2}$ to the right

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